

3.5

De Kettingregel en Parametervergelijkingen

Hoe leiden we $(g \circ f)$ af, zoals bijvoorbeeld $\cos(x^2+1)$ en $(3x^2+1)^2$?

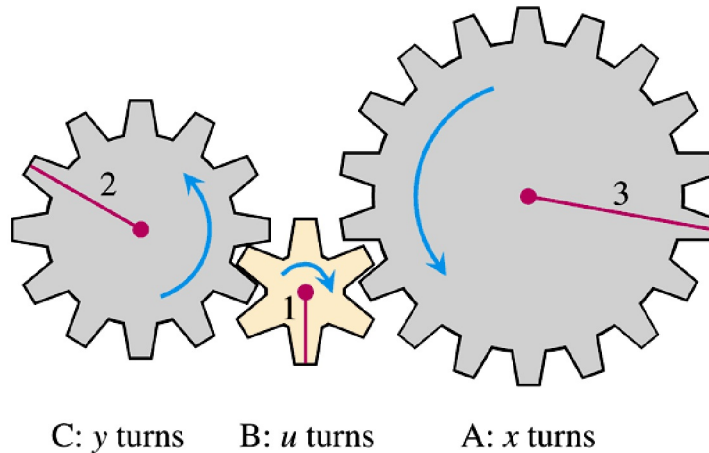


FIGURE 3.26 When gear A makes x turns, gear B makes u turns and gear C makes y turns. By comparing circumferences or counting teeth, we see that $y = u/2$ (C turns one-half turn for each B turn) and $u = 3x$ (B turns three times for A's one), so $y = 3x/2$. Thus, $dy/dx = 3/2 = (1/2)(3) = (dy/du)(du/dx)$.

$$\cos(x^2 + 1)$$

$$x \rightarrow x^2 + 1 = u \rightarrow \cos u$$

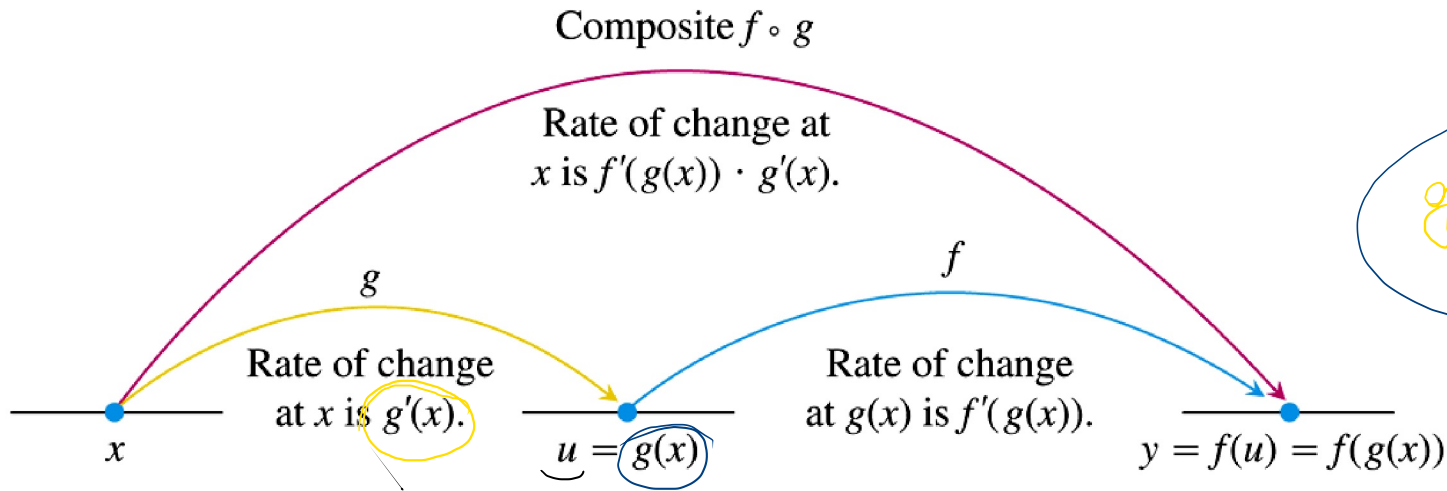
$$f(x) = x^2 + 1$$

$$g(x) = \cos x$$

$$g \circ f(x) = \cos(x^2 + 1)$$

$$g(f(x)) = \cos(x^2 + 1)$$

Kettingregel



$$g'(x) f'(g(x))$$

Stelling

Indien $g(x)$ differentieerbaar is in x en $f(u)$ differentieerbaar in $g(x)$ dan is de samengestelde

functie $y(x) = (f \circ g)(x) = f(g(x))$ differentieerbaar in x :

$$y'(x) = (f \circ g)'(x) = f'(g(x))g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Notatie van Leibniz:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

in x in x in $u(x)$

Voorbeelden:

Outside-inside regel !

$$(\sin(x))^5 = \sin^5 x$$

1. Zoek de raaklijn aan $y = \sin^5 x$ voor $x = \pi/3$?

$$x \rightarrow \sin(x) = u \rightarrow u^5$$

$\cos(x)$ $5u^4$

$$(\sin^5 x)' = \cos(x) 5 \sin^4 x$$

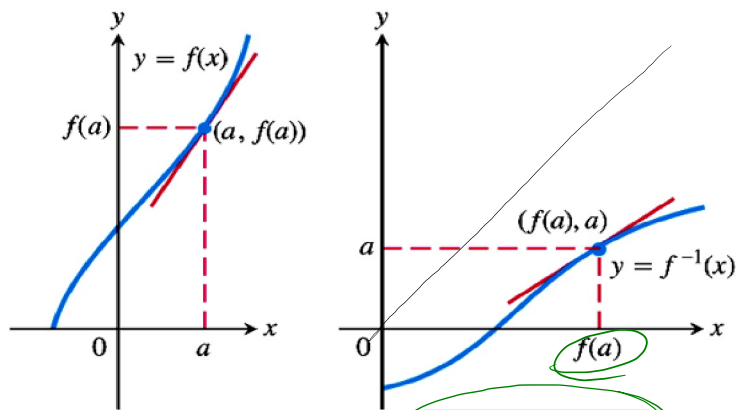
2. Bereken $\frac{d(\sqrt{\sin(2x)})}{dx} = 2 \cos(2x) \frac{1}{2 \sqrt{\sin(2x)}} = \frac{\cos(2x)}{\sqrt{\sin(2x)}}$

$$x \rightarrow 2x = u \rightarrow \sin u = v \rightarrow \sqrt{v} = v^{1/2}$$

$\downarrow$ $\downarrow$ $\downarrow$
 2 $\cos u$ $\frac{1}{2} v^{-1/2} = \frac{1}{2\sqrt{v}}$

Gevolg: De afgeleide van de inverse functie (Zie ook paragraaf 7.1 in boek)

$$\frac{df}{dx}\bigg|_a$$



The slopes are reciprocal:

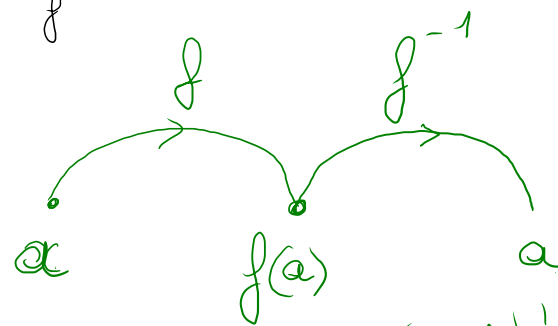
$$\frac{df^{-1}}{dx}\bigg|_{f(a)} = \frac{1}{\frac{df}{dx}\bigg|_a}$$

$$f \rightarrow f^{-1}$$

$$x \xrightarrow{f} f(x) \xrightarrow{f^{-1}} x$$

$$f^{-1} \circ f(x) = x$$

$$(f^{-1} \circ f)' = 1$$



$$f'(a) \cdot (f^{-1})'(f(a)) = 1$$

$$(f^{-1})'\bigg|_{f(a)} = \frac{1}{f'(a)}$$

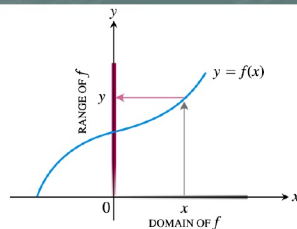
Inverse functie



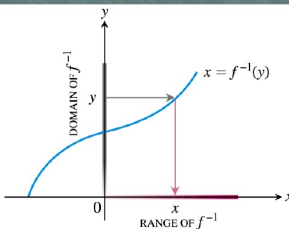
x en y omwisselen



Grafiek spiegelen tov
bissectrice van eerste kwadrant

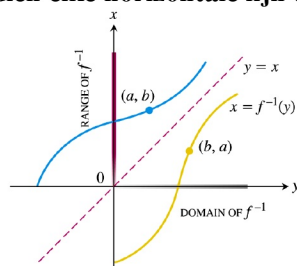
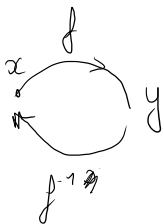


(a) To find the value of f at x , we start at x , go up to the curve, and then over to the y -axis.

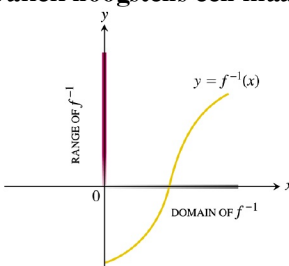


(b) The graph of f is already the graph of f^{-1} , but with x and y interchanged. To find the x that gave y , we start at y and go over to the curve and down to the x -axis. The domain of f^{-1} is the range of f . The range of f^{-1} is the domain of f .

Een inverse functie bestaat indien elke horizontale lijn de grafiek hoogstens één maal snijdt.

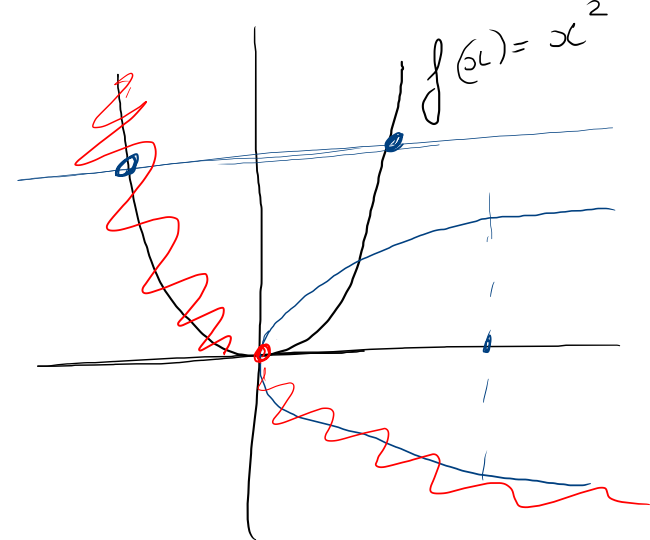


(c) To draw the graph of f^{-1} in the more usual way, we reflect the system in the line $y = x$.

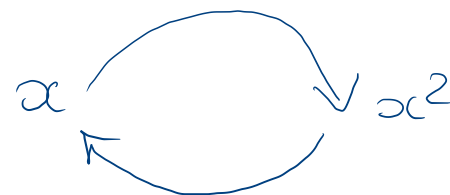


(d) Then we interchange the letters x and y . We now have a normal-looking graph of f^{-1} as a function of x .

FIGURE 7.2 Determining the graph of $y = f^{-1}(x)$ from the graph of $y = f(x)$.



$$f^{-1}(x) = \sqrt{x}$$



Voor de functie $y=x^2$ bestaat er enkel een inverse indien we het domein beperken tot $x \geq 0$

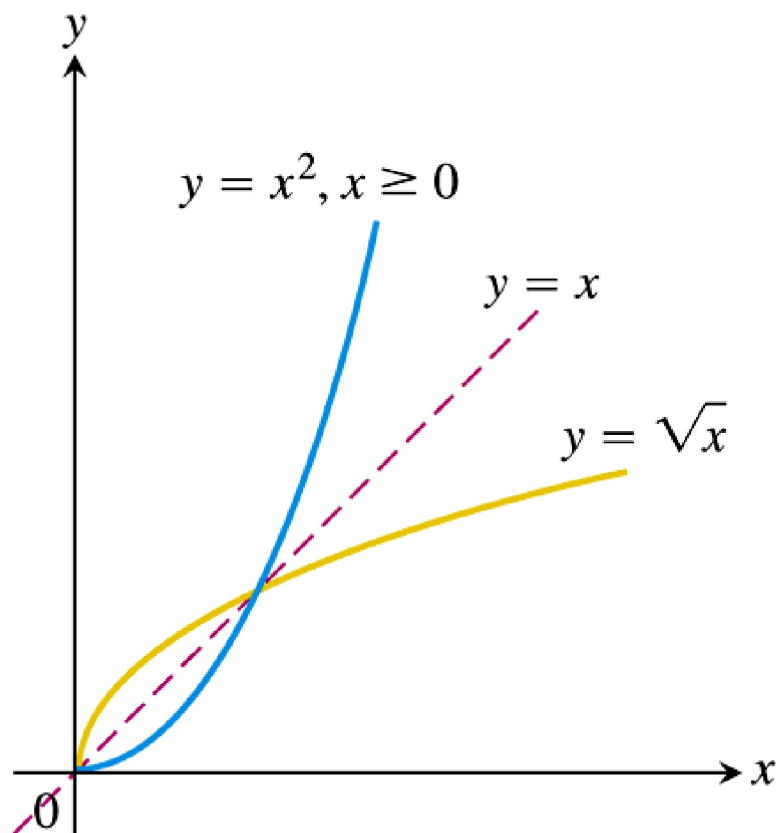


FIGURE 7.4 The functions $y = \sqrt{x}$ and $y = x^2, x \geq 0$, are inverses of one another (Example 3).

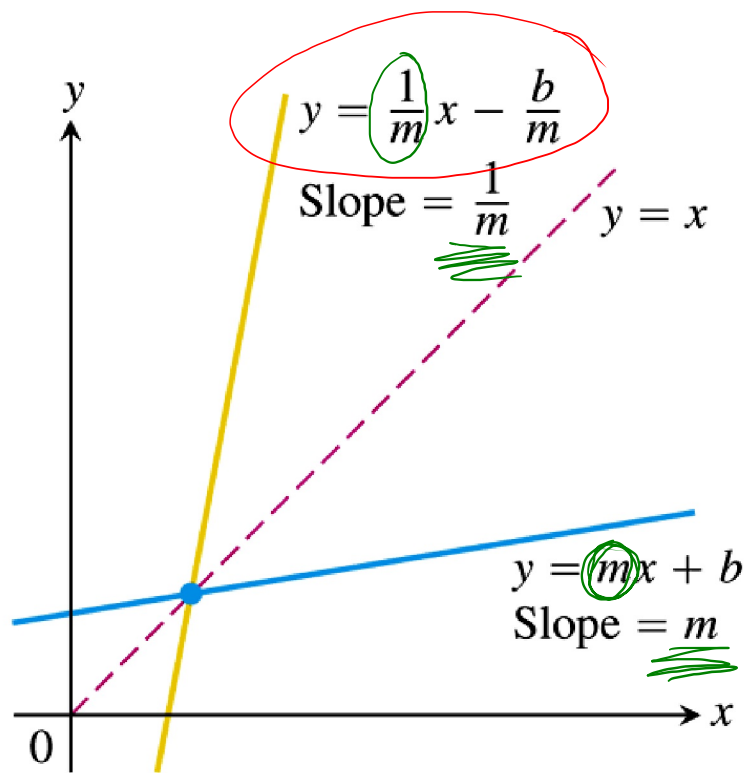
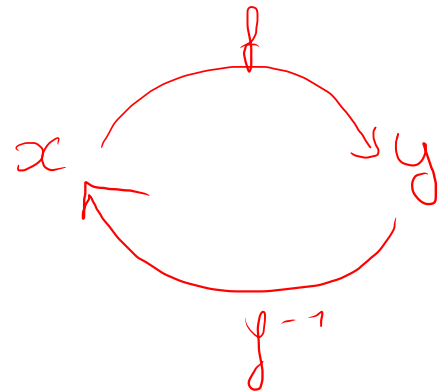


FIGURE 7.5 The slopes of nonvertical lines reflected through the line $y = x$ are reciprocals of each other.

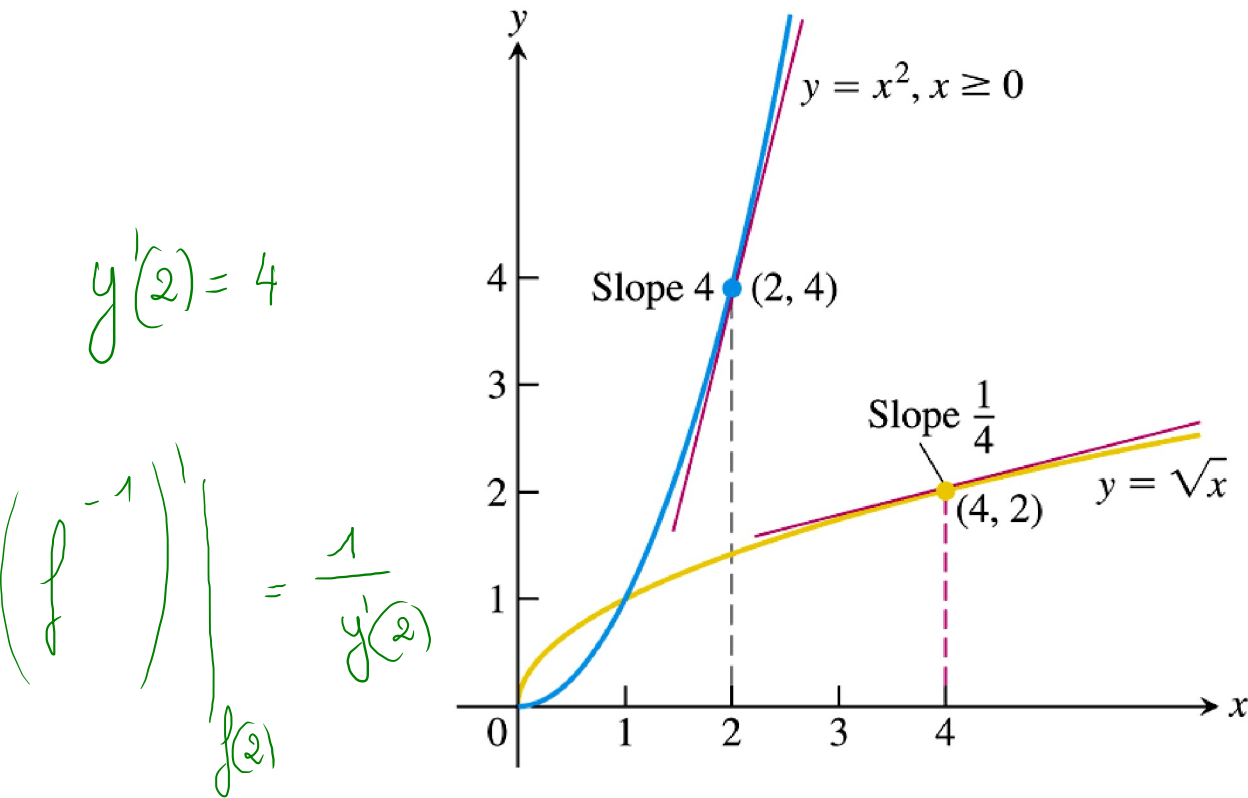
$$y = mx + b$$



$$mx = y - b$$

$$x = \frac{1}{m}y - \frac{b}{m}$$

Voorbeeld van afgeleide van een inverse functie



De raaklijn van de functie $f(x)=x^2$ in het punt $(2,4)$ met rico 4 wordt na spiegeling de raaklijn van de inverse functie in het punt $(4,2)$ met rico $\frac{1}{4}$.

Dit bevestigt de regel:

$$\left. \frac{df^{-1}}{dx} \right|_{f(a)} = \frac{1}{\left. \frac{df}{dx} \right|_a}$$

waarbij $a=2$ en $f(a)=4$

FIGURE 7.7 The derivative of $f^{-1}(x) = \sqrt{x}$ at the point $(4, 2)$ is the reciprocal of the derivative of $f(x) = x^2$ at $(2, 4)$ (Example 4).

Vb:

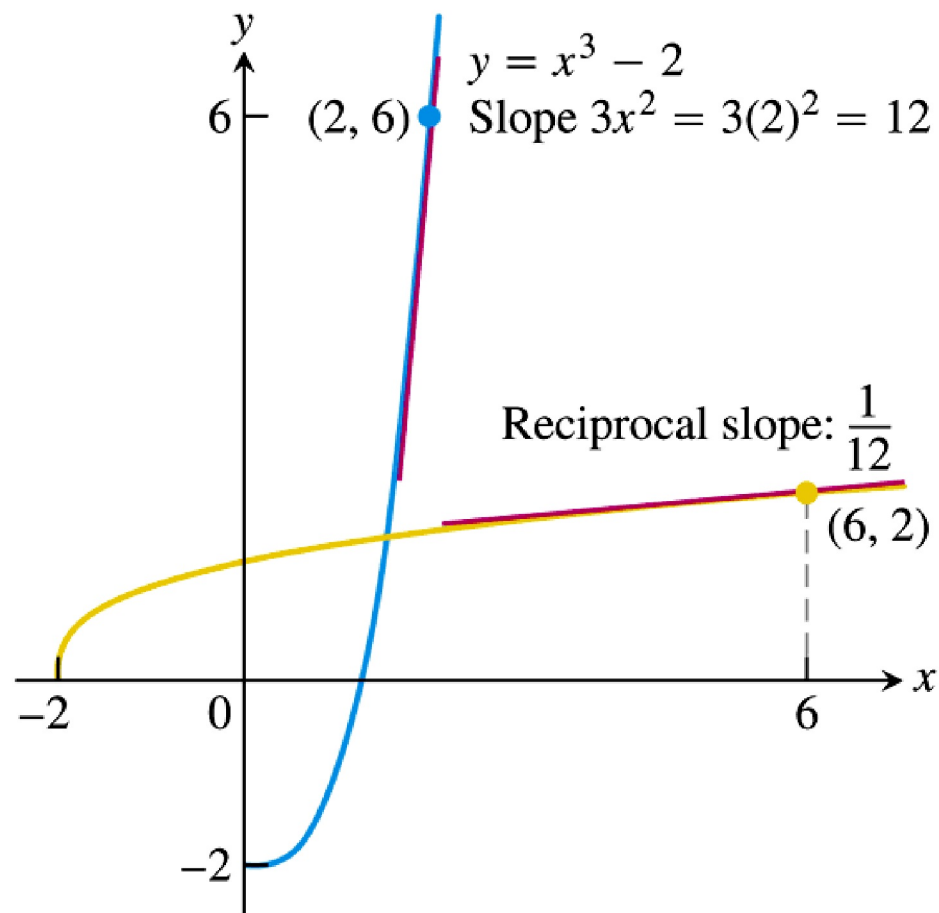


FIGURE 7.8 The derivative of $f(x) = x^3 - 2$ at $x = 2$ tells us the derivative of f^{-1} at $x = 6$ (Example 5).

!!! De trigonometrische formules voor afgeleiden gelden enkel indien we in radialen werken !!!!

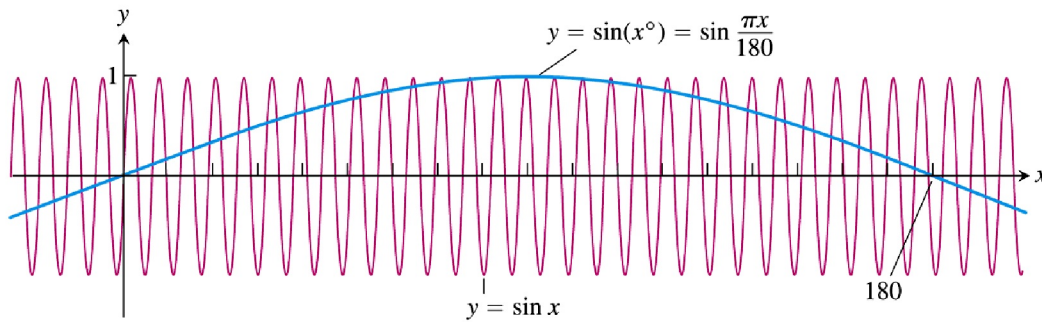


FIGURE 3.28 $\sin(x^\circ)$ oscillates only $\pi/180$ times as often as $\sin x$ oscillates. Its maximum slope is $\pi/180$ at $x = 0$ (Example 8).

Helling van grafiek
Veel minder steil

$$\frac{d}{dx} \sin(x^\circ) = \frac{d}{dx} \sin\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos(x^\circ)$$

$$\begin{aligned} 180^\circ &\rightarrow \pi \text{ rad} \\ 1^\circ &\rightarrow \frac{\pi}{180} \text{ rad} \\ x^\circ &\rightarrow \frac{\pi x}{180} \text{ rad} \end{aligned}$$

$$\begin{aligned} x &\rightarrow \frac{\pi x}{180} = u && \rightarrow \sin u \\ &\downarrow && \downarrow \\ \frac{\pi}{180} &&& \cos u \end{aligned}$$

Parametervoorstelling van een 2-dimensionale kromme:

x en y zijn functies van een parameter t :

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases} \text{ voor } a \leq t \leq b$$

De punten $(x,y)=(f(t),g(t))$ vormen de kromme.

$(f(a),g(a))$ is het startpunt en
 $(f(b),g(b))$ is het eindpunt
van de kromme

Met een parametervergelijking kun
je krommes beschrijven die geen
functies zijn

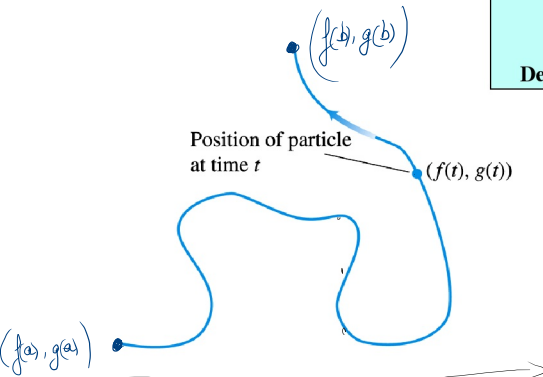
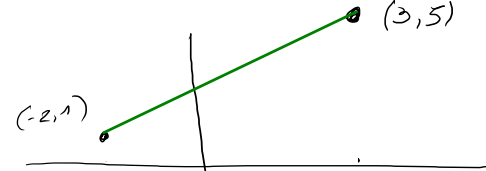


FIGURE 3.29 The path traced by a particle moving in the xy -plane is not always the graph of a function of x or a function of y .

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Voorbeeld: Zoek het lijnsegment
dat de punten $(-2,1)$ en $(3,5)$ verbindt



$$y - 1 = \frac{5 - 1}{3 - (-2)} (x + 2)$$

$$y - 1 = \frac{4}{5} (x + 2)$$

$$\textcircled{1} \quad \frac{y-1}{4} = \frac{x+2}{5} = t$$

$$\begin{cases} x = 5t - 2 \\ y = 4t + 1 \end{cases} \quad 0 \leq t \leq 1$$

$$\textcircled{2} \quad y = \frac{4}{5}x + \frac{13}{5}$$

$$\begin{cases} x = t \\ y = \frac{4}{5}t + \frac{13}{5} \end{cases} \quad -2 \leq t \leq 3$$

Parametervoorstelling van een cirkel met straal 1

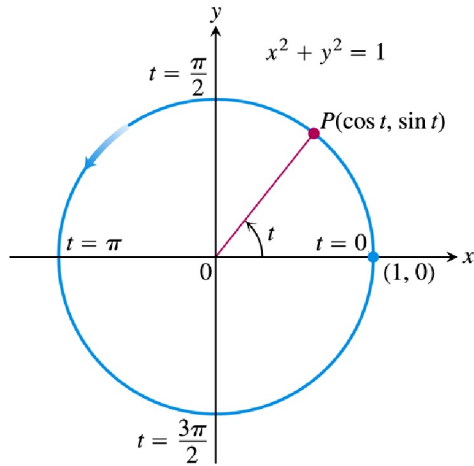


FIGURE 3.30 The equations $x = \cos t$ and $y = \sin t$ describe motion on the circle $x^2 + y^2 = 1$. The arrow shows the direction of increasing t (Example 9).

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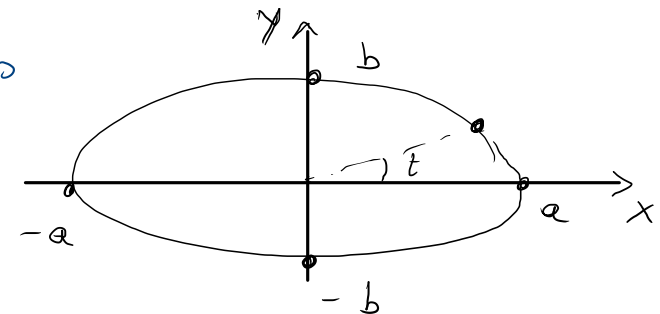
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cirkel met straal a

$$\begin{cases} x(t) = a \cos(t) \\ y(t) = a \sin(t) \end{cases}$$

$$0 \leq t < 2\pi$$

ellips

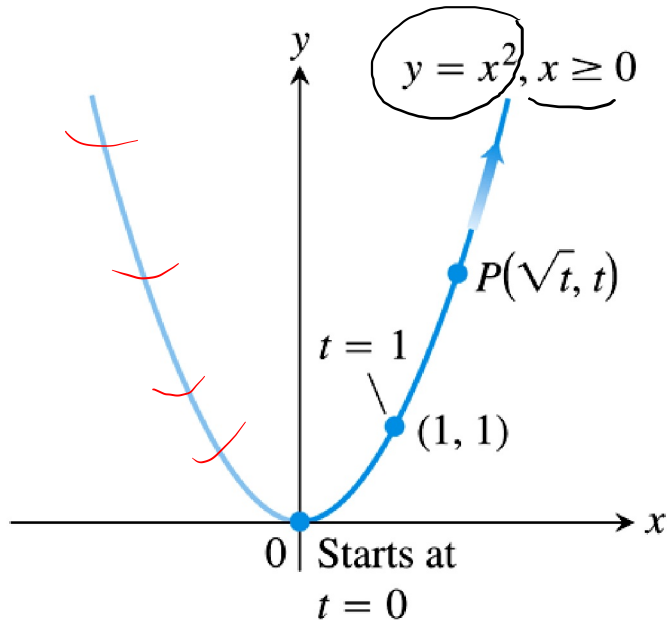


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\begin{cases} x(t) = a \cos t \\ y(t) = b \sin t \end{cases}$$

$$0 \leq t < 2\pi$$

Parametervoorstelling van de parabool $y=x^2$ voor positieve x-waarden



$$\begin{cases} x(t) = \sqrt{t} \\ y(t) = t \end{cases} \quad t \geq 0$$

of

$$\begin{cases} x(t) = t \\ y(t) = t^2 \end{cases} \quad t \geq 0$$

FIGURE 3.31 The equations $x = \sqrt{t}$ and $y = t$ and the interval $t \geq 0$ describe the motion of a particle that traces the right-hand half of the parabola $y = x^2$ (Example 10).

! Interval van t bij 2 verschillende parametervoorstellingen van dezelfde kromme kan verschillen !

→ V_b van lynstuk

Helling van de raaklijn aan een parametrische kromme

Parametric Formula for dy/dx

If all three derivatives exist and $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad (2)$$

Bewijs: Kettingregel $\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$

Voorbeeld: parabool

$$\begin{cases} x(t) = t \\ y(t) = t^2 \end{cases} \longrightarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t}{1} = 2t$$

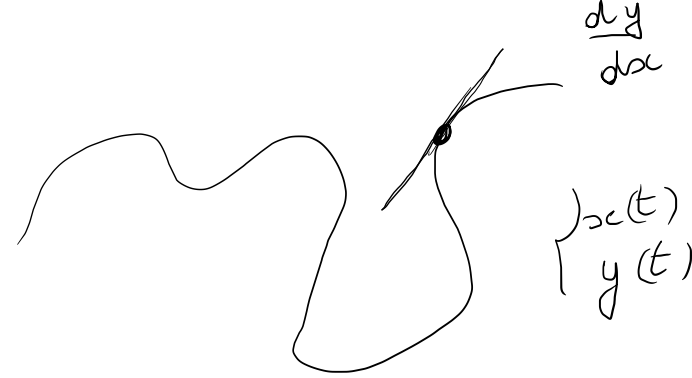
Parametric Formula for d^2y/dx^2

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(y') = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$$

NB: y' uitrekenen in functie van t mbv 1^{ste} formule

Vb: parabool

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t}{1} = 2t \longrightarrow \frac{d^2y}{dx^2} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}} = \frac{2}{1} = 2$$



$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} y' = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$$

Parametric Formula for d^2y/dx^2

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(y') = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$$

NB: y' uitrekenen in functie van t mbv 1^{ste} formule

Vb: parabool

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t}{1} = 2t \quad \longrightarrow \quad \frac{d^2y}{dx^2} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}} = \frac{2}{1} = 2$$

- 1) Waar landt de cargo ?
- 2) Geef een cartesische vergelijking van de baan van de cargo.
- 3) Hoe snel daalt de hoogte met x bij landing ?

$$\begin{cases} x = 120t \\ y = -16t^2 + 500 \end{cases}$$

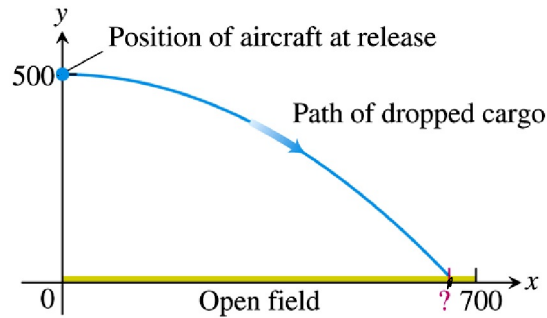


FIGURE 3.32 The path of the dropped cargo of supplies in Example 15.

$$\begin{aligned} 1) \quad y(t_e) &= 0 \\ -16t_e^2 + 500 &= 0 \\ t_e^2 &= \frac{500}{16} = \frac{125}{4} \\ t_e &= \sqrt{\frac{125}{4}} = \frac{\sqrt{125}}{2} \end{aligned}$$

$$x(t_e) = 120 \times \frac{\sqrt{125}}{2}$$

$$2) \quad t = \frac{x}{120}$$

$$y = -16 \frac{x^2}{120^2} + 500$$

$$3) \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-32t}{120}$$

$$\left. \frac{dy}{dx} \right|_{t_e} = -\frac{32}{120} t_e$$

NB: in "feet"

Standard Parametrizations and Derivative Rules

CIRCLE $x^2 + y^2 = a^2$:

$$x = a \cos t$$

$$y = a \sin t$$

$$0 \leq t \leq 2\pi$$

FUNCTION

$$y = f(x):$$

$$\left\{ \begin{array}{l} x = t \\ y = f(t) \end{array} \right.$$

ELLIPSE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

$$x = a \cos t$$

$$y = b \sin t$$

$$0 \leq t \leq 2\pi$$

DERIVATIVES

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}$$

3.6

Impliciete Afgeleiden

Tot nu toe: afgeleide van functie gedefinieerd indien functie gegeven wordt als

- $y = f(x)$
- Een parametervergelijking $\begin{cases} x = f(t) \\ y = g(t) \end{cases}$



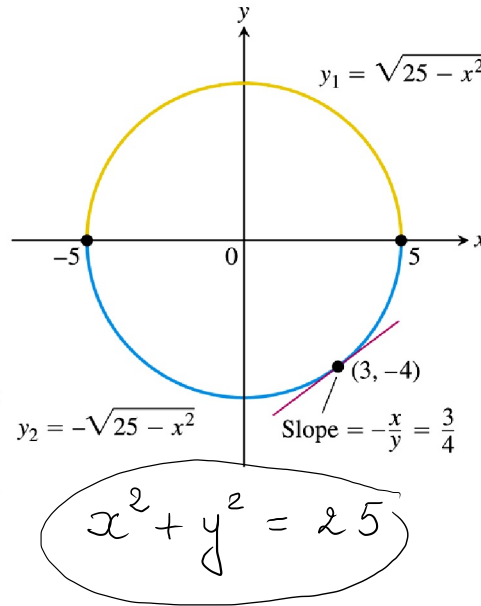
Nu : **functie impliciet gedefinieerd**

bv: $x^2 + y^2 = 25$

Afgeleide van zo'n functie in een punt ?



- Differentieer linker- en rechterlid naar x waarbij je y als een differentieerbare functie naar x beschouwt
- Los de bekomen uitdrukking op naar dy/dx



$$x \rightarrow \begin{matrix} (y(x))^2 \\ y(x) = u \rightarrow u^2 \\ \quad \quad \quad \nwarrow 2u \\ y' \end{matrix}$$

1. Samenstelling van 2 functies y_1 en y_2

$$y_1' = \frac{\cancel{2x}}{2\sqrt{25-x^2}} = \frac{-x}{y_1}$$

$$y_2' = \frac{\cancel{2x}}{2\sqrt{25-x^2}} = \frac{x}{-y_2}$$

2. parametervergelijking

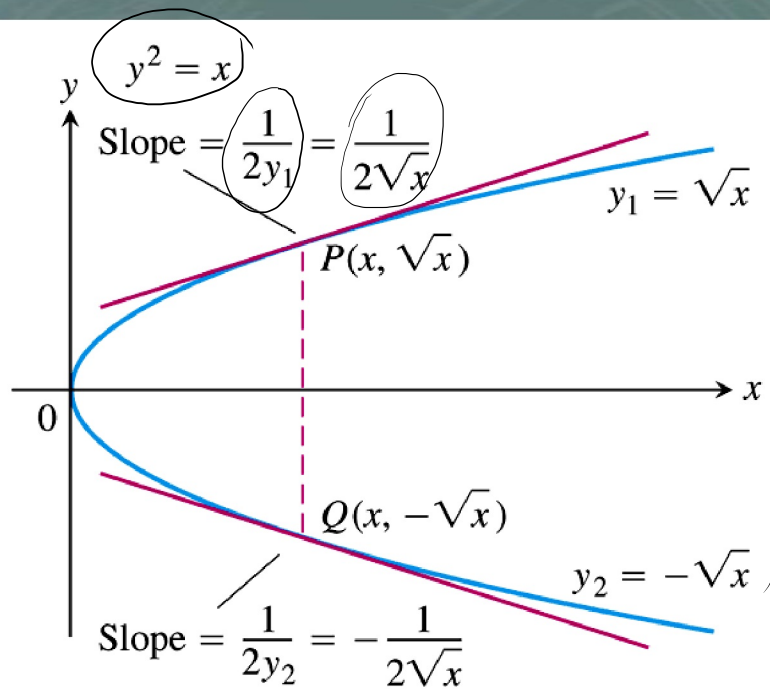
$$\begin{cases} x(t) = 5 \cos t \\ y(t) = 5 \sin t \end{cases}$$

$$\frac{dy}{dx} = \frac{5 \cos t}{-5 \sin t} = \frac{x}{-y}$$

3. impliciet afleiden

$$\cancel{2x} + \cancel{2y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$



$$y^2 = x$$

$$2y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2y}$$

FIGURE 3.37 The equation $y^2 - x = 0$, or $y^2 = x$ as it is usually written, defines two differentiable functions of x on the interval $x \geq 0$. Example 1 shows how to find the derivatives of these functions without solving the equation $y^2 = x$ for y .

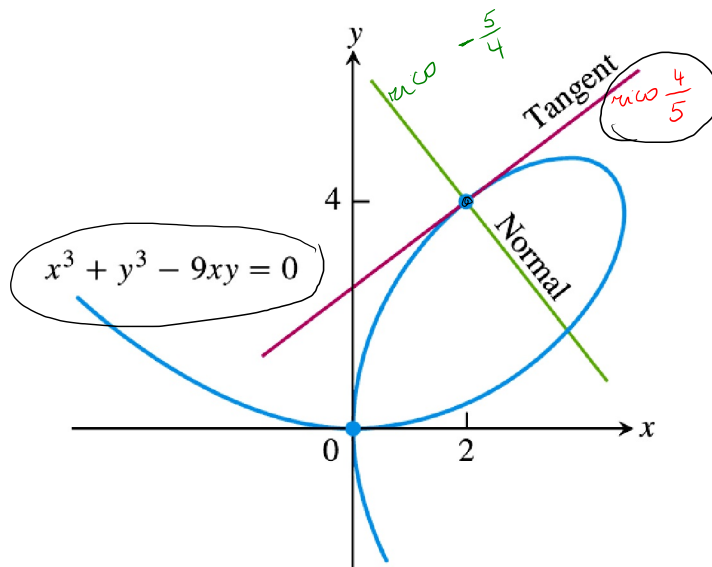
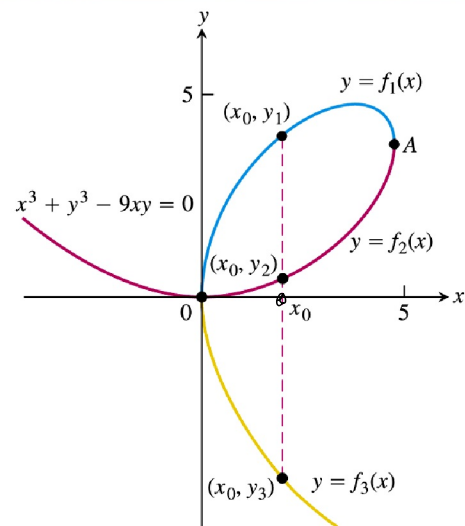


FIGURE 3.41 Example 4 shows how to find equations for the tangent and normal to the folium of Descartes at $(2, 4)$.

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$$x^3 + y^3 - 9xy = 0$$

$$3x^2 + 3y^2 \frac{dy}{dx} - 9(y + x \frac{dy}{dx}) = 0$$

$$\frac{dy}{dx} (3y^2 - 9x) = -3x^2 + 9y$$

$$\frac{dy}{dx} = \frac{3y^2 - 3x^2}{3y^2 - 9x}$$

$$\frac{dy}{dx} \bigg|_{(2,4)} = \frac{12 - 4}{16 - 6} = \frac{8}{10} = \frac{4}{5}$$

$$-\frac{1}{\frac{4}{5}} = -\frac{5}{4}$$

FIGURE 3.38 The curve $x^3 + y^3 - 9xy = 0$ is not the graph of any one function of x . The curve can, however, be divided into separate arcs that are the graphs of functions of x . This particular curve, called a *folium*, dates to Descartes in 1638.

Bereken de helling van de onderstaande grafiek in elk punt (x,y)

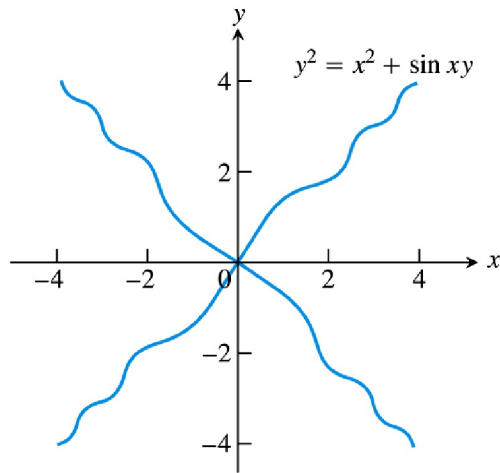


FIGURE 3.39 The graph of $y^2 = x^2 + \sin xy$ in Example 3. The example shows how to find slopes on this implicitly defined curve.

$$y^2 = x^2 + \sin(xy)$$

$$2y \frac{dy}{dx} = 2x + \cos(xy) \left(y + x \frac{dy}{dx} \right)$$

$$\left(2y - \cos(xy) x \right) \frac{dy}{dx} = 2x + y \cos(xy)$$

$$\frac{dy}{dx} =$$

Analoog: Hogere orde afgeleide

$$\text{Vb. } 2x^3 - 3y^2 = 8 \longrightarrow y' = \frac{x^2}{y} \quad \text{en} \quad y'' = \frac{2x}{y} - \frac{x^4}{y^3}$$

Via impliciet afleiden kan men de onderstaande uitbreiding van de reeds gekende formule bewijzen

THEOREM 4 Power Rule for Rational Powers

If p/q is a rational number, then $x^{p/q}$ is differentiable at every interior point of the domain of $x^{(p/q)-1}$, and

$$\frac{d}{dx} x^{p/q} = \frac{p}{q} x^{(p/q)-1}.$$

Bewijs in de les

$$2x^3 - 3y^2 = 8$$

$$6x^2 - 6y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x^2}{y}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{2xy - x^2 \frac{dy}{dx}}{y^2} \\ &= \frac{2xy - x^2 \frac{x^2}{y}}{y^2} \\ &= \frac{2x}{y} - \frac{x^4}{y^3} \end{aligned}$$

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$$\left(\frac{y}{x} \right)^q = \left(x^{p/q} \right)^q = x^p$$

$$y^q = x^p$$

$$q y^{q-1} \frac{dy}{dx} = p x^{p-1}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{p}{q} x^{p-1} y^{1-q} \\ &= \frac{p}{q} x^{p-1} \left(x^{p/q} \right)^{1-q} \\ &= \frac{p}{q} x^{\frac{(p-1)q + p(1-q)}{q}} \end{aligned}$$

$$\begin{aligned} &= \frac{p}{q} x^{\frac{p-q}{q}} \\ &= \frac{p}{q} x^{\frac{p}{q} - 1} \end{aligned}$$

3.7

Vraagstukken met snelheden

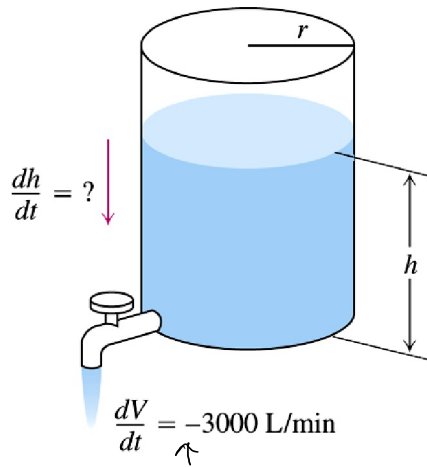


FIGURE 3.42 The rate of change of fluid volume in a cylindrical tank is related to the rate of change of fluid level in the tank (Example 1).

$$\begin{aligned}
 V &= \pi R^2 h \quad 1000 \frac{\text{l}}{\text{m}^3} \\
 -3000 \frac{\text{l}}{\text{min}} &= \frac{dV}{dt} = \pi R^2 \frac{dh}{dt} \quad 1000 \frac{\text{l}}{\text{m}^3} \\
 \frac{dh}{dt} &= \frac{-3000 \frac{\text{l}}{\text{min}}}{\pi R^2 \quad 1000 \frac{\text{l}}{\text{m}^3}} \\
 &= \frac{-3}{\pi R^2} \frac{\text{m}^3}{\text{min}}
 \end{aligned}$$

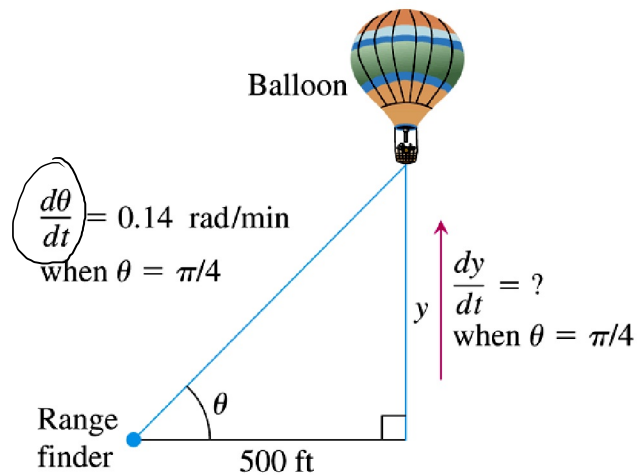


FIGURE 3.43 The rate of change of the balloon's height is related to the rate of change of the angle the range finder makes with the ground (Example 2).

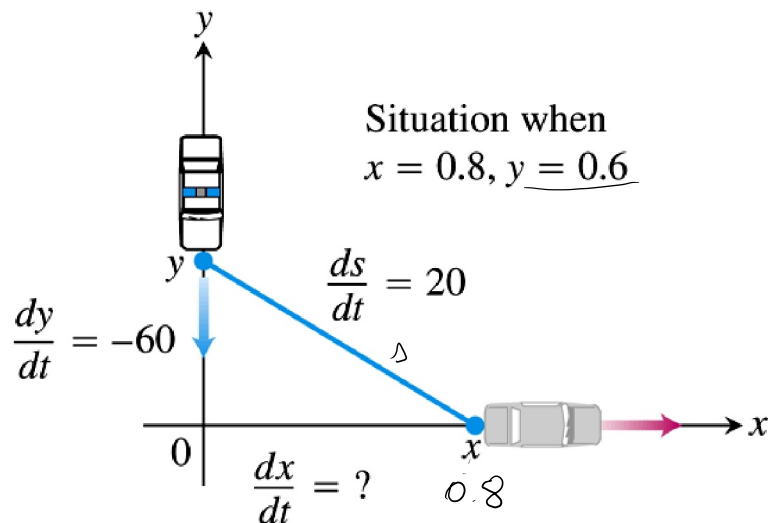
$$\tan \theta = \frac{y}{500}$$

$$\frac{\frac{dy}{dt}}{\cos^2 \theta} = \frac{1}{500} \frac{dy}{dt}$$

$$\left. \frac{dy}{dt} \right|_{\theta = \frac{\pi}{4}} = \frac{500 \text{ ft} \cdot 0.14 \frac{\text{rad}}{\text{min}}}{\cos^2 \left(\frac{\pi}{4} \right)}$$

$$= 1000 \text{ ft} \cdot 0.14 / \text{min}$$

$$= 140 \frac{\text{ft}}{\text{min}}$$



$$x^2 + y^2 = s^2$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2s \frac{ds}{dt}$$

FIGURE 3.44 The speed of the car is related to the speed of the police cruiser and the rate of change of the distance between them (Example 3).

NB: 3.8: Differentialen NIET !!!